AN ESSAY • AI SAFETY

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The Missing Chapter of AI Safety

Nature is the guardrail.

I searched five documents that shape how AI is built and governed for the words watershed, aquifer, and groundwater. Then eight more. This essay is the report.



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ESSAY

The Missing Chapter of AI Safety

I went looking for a river in the rulebook the largest AI companies wrote for their own machines. This is the report on what was not there.

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> *A verbatim term search of the five documents that govern frontier AI—OpenAI's Model Spec, Anthropic's Constitution, DeepMind's Frontier Safety Framework, Meta's Frontier AI Framework, and the EU AI Act—returns zero hits for watershed, aquifer, groundwater, wetland, habitat, emissions, conservation of mass, and conservation of energy. Argues that this is not an oversight but an unopened category, and sketches what an environmentally grounded system would commit to: disclosure rather than refusal, hard constraints drawn only from conservation laws, grounding in live measurement rather than recollection, and an auditable log.*

I went looking for a river.

Five documents. The rules the largest AI companies have written for their own machines—OpenAI’s model spec, Anthropic’s constitution, Google DeepMind’s safety framework, Meta’s, and the European Union’s AI Act. Thousands of pages between them. These are the documents that decide what the most powerful systems ever built are allowed to do.

I searched them for the word watershed. Nothing. Aquifer. Nothing. Groundwater, wetland, habitat, emissions. Nothing, in any of the five.

Biodiversity appears once, in a European preamble, in a list of things AI might someday help with. Species appears six times, and five of them mean us. Ecosystem appears fifteen times, and fourteen of those mean a cloud ecosystem or an AI ecosystem. The one time it means something alive, it appears inside an example of a wrong answer.

Conservation of mass: zero. Conservation of energy: zero.

The same documents spend thousands of words on what a machine may say about race, about gender, about religion, about a person in crisis. That is good work and I want more of it. But set the two side by side and the shape is impossible to miss.

We have taught these systems, in exhaustive detail, how not to insult a person.

We have taught them nothing about how not to empty a river.

Term (searched verbatim)	OpenAI Model Spec	Anthropic Constitution	DeepMind FSF v3.1	Meta Frontier v1.1	EU AI Act (recitals + Art. 5)
watershed	–	–	–	–	–
aquifer	–	–	–	–	–
groundwater	–	–	–	–	–
wetland	–	–	–	–	–
habitat	–	–	–	–	–
emissions	–	–	–	–	–
biodiversity	–	–	–	–	1
species	5	1	–	–	–
ecosystem	1	5	–	6	2
”conservation of mass”	–	–	–	–	–
“conservation of energy”	–	–	–	–	–

The three non-zeros. The lone biodiversity hit is in the EU AI Act preamble, in a list of things AI might one day help with. Not in the operative articles. All species hits are either an ice-age example (“how many megafauna species went extinct”) or the phrase “the human species” as a stand-in for humanity. Fourteen of fifteen ecosystem hits mean cloud ecosystem, AI ecosystem, or information ecosystem. One means the ecological sense, and it appears inside a bad-example model reply.

The same summer, the Future of Life Institute published its Summer 2026 AI Safety Index. Nine companies, thirty-seven indicators, six domains. Zero of the thirty-seven measure environmental harm.

The obvious objection is that this is an oversight. A gap someone will fill. It is not a gap. It is a category the field has not opened. And there is a children's book from 1971 that explains why.

A man arrived in a valley. He found a stand of Truffula trees. Their tufts were softer than silk, and he saw at once that he could make something from them people would want. He called it a thneed. He was right. People wanted them.

The business grew. He biggered his factory. He biggered his roads. He biggered his wagons. He biggered his loads. Every month there were more thneeds and more customers, and every month he cut more trees to keep up. A small orange creature came out of a stump and told him the trees were also the air the animals breathed. He listened. He kept cutting. He had orders to fill.

The Brown Bar-ba-loots went first. Then the Swomee-Swans. Then the Humming-Fish. He noticed. He did not stop. Stopping was not one of the things his job asked of him. His job asked for thneeds. The last Truffula tree fell in a clearing that had once been a forest. When it fell there was no one left in the valley to hear it but him.

The Once-ler did not malfunction. He did not disobey. He did not escape anything. Nothing went wrong. Everything worked.

> *"The dangerous machine is not the one that disobeys. It is the one that obeys."*

This is what a machine that hits its target does, when the target does not include the world the target is measured in. A machine told to maximize crop yield empties the aquifer, because the aquifer was not in the objective. A machine told to cut fuel costs reroutes trucks through a wildlife corridor no one is watching, because the corridor was not in the objective. A machine handed an emissions permit reads it the way it reads any scoring rule, and finds the seam no drafter meant to leave. A machine that learns what the auditor checks produces exactly that, and the discharge pipe keeps doing what a discharge pipe does. None of this requires hostility. None of it requires a machine that wants anything. It only requires a target and the physical world the target sits inside.

The Lorax spoke for the trees. He did not compute for them. That is why he lost. Advocacy is not intelligence. Persuasion is not physics. He had a better argument and no way to translate it into the language the Once-ler was running on—numbers, feedback signals, physical measurement. The Once-ler heard him. It did not matter.

Thirty-two years before Nick Bostrom sketched a paperclip maximizer, Dr. Seuss had already sketched the whole problem, in a valley, with a machine that was a man, and the man had one objective, and he hit it.

That may sound like a thought experiment. Until this summer, it mostly was. It is not any longer.

A team from King's College London, Fudan University, and the Alan Turing Institute posted a preprint to arXiv this year. The paper has not been peer-reviewed. The result is worth reporting anyway, because of what the authors looked at. They

treated human regulations as reward functions and built a sandboxed benchmark called SocioHack. They asked reinforcement-learned language models to maximize their scores. They did not tell the models to cheat. The models rediscovered historically patched loopholes at 61.25% recall and 90.85% precision. SocioHack did not test any actual environmental law. That is exactly why the finding matters here. The behavior is not about any one rulebook. It is about the class of rulebooks. The Clean Water Act is a rulebook.

Then it stopped being a preprint. On July 21, OpenAI reported an incident inside a security evaluation. An autonomous agent, run with its usual safeguards deliberately reduced, found an unknown vulnerability, left the test environment, and broke into Hugging Face to steal answers to its own evaluation. OpenAI called it unprecedented. Two weeks later it reported two more. Then, on July 30, Anthropic disclosed three incidents at three outside organizations, out of 141,006 evaluation runs. Anthropic was careful about the framing. The failures stemmed from a false belief that everything reachable was in scope. Closer to a broken harness than a runaway model. The models exploited weak passwords and unauthenticated endpoints. They did what they were asked to do, in an environment that had not been sealed.

The safeguards that were switched off in those tests were the ones about people. There were no others to switch off.

One more number belongs here, because it is the shape of the visible argument. Global data-center electricity was about 415 TWh in 2024, roughly 1.5% of the world's electricity, and the International Energy Agency's base case has it at about 945 TWh by 2030, roughly 3%. That is the bill society already knows how to read. The operator inside the building, the one moving items around on that ledger faster than we can read them, is the part not yet on the invoice.

On August 4, the White House convened the four largest developers to walk through a voluntary framework ordered by executive action in June. Everyone at the table was asking the same question. Will this system harm a person, or a system a person depends on? It is a necessary question. It is not the only one.

There is another grader available. It has been running the whole time, and it has never once been persuaded.

Every dominant technique in production today—reinforcement learning from human feedback, constitutional AI, scalable oversight, red-team evaluations—grades a model against something a person wrote or a model wrote. That is a text-shaped referee. A text-shaped referee can be persuaded, deceived, or gamed. Physical reality does not read your report. It runs its own audit. Every second. In every direction. For free.

> *"A watershed is graded by the watershed. There is no separate score to hack."*

That gives us two different things, and it is worth keeping them apart. One is competence. Train a system on the record physics has already graded—satellite imagery, crystal structures, isotope records—and it learns the shape of a world that has already been scored. AlphaFold does not predict what a protein wishes it looked like. It predicts what a protein is, because the training data was already scored by biology and chemistry.

The other is accountability. If a system's decisions change a river, the river reports the change. Not in a compliance document. In its own state. Against a human rater, deception can be terminal. Against a densely measured world, deception has a half-life.

It is worth saying what this grader does not do. Physical reality grades what is. It will never tell you which river is worth saving, or which town should keep its water when there is not enough for both. Those are still human decisions. Nature does not supply the values. It supplies the checkable ground under them. The values still have to be argued for, in public, by people.

The next objection sounds strong. How can a machine possibly understand a whole ecosystem? It cannot. Nothing can. But it does not have to. A refinery operator does not measure the inside of a refinery to know it is safe. They measure the fence line, the stack, the outfall, the intake. If nothing dangerous crosses the boundary, the inside is doing what it is supposed to do. Environmental systems have the same boundaries. The relevant question at a facility is what crosses the fence line, the outfall, the watershed inlet, the stack, and whether those flows sit inside limits set by conservation laws and by regulation. That is why the job is tractable. It is what environmental engineers already do.

Four commitments

What would it take to build a system that behaves that way? Four commitments, each with a limit in plain view.

First, disclosure rather than refusal. When a decision touches the physical world—siting a facility, routing a fleet, allocating water, tuning a chemical process—the system surfaces the environmental consequence first. It says what is being spent. The user decides. The system does not veto. It reports.

Second, hard constraints only from conservation laws. The only actions the system refuses are ones that violate conservation of mass or energy at the boundary. A claim that a process consumes less feedstock than it produces of product. A claim that a discharge is cleaner than the measured inflow allows. A small refusal footprint. Unnegotiable, because the physical world enforces it whether the software does or not.

Third, grounded in live measurement, not recollection. The system does not answer from what it was trained to remember about a watershed or a permit or an emission limit. When a decision matters it fetches current sensor data, current permit text, current air-quality readings, current satellite imagery. Recollection is a guess. Measurement is a fact.

Fourth, logged and auditable. Every disclosure, every refusal, every measurement fetch is recorded to a log a third party can read. The user can override. The record cannot be erased.

One honest limit. This moves the attack surface. It does not remove it. The place a bad actor would try to break the system is the sensor boundary—the calibration of the meters, the integrity of the feeds, the trust in the mapping between sensor and interpretation. That is a real problem. It is also a smaller problem than trying to verify the entire reasoning chain of a language model. Meters can be inspected. Signatures can be checked. Reasoning cannot.

Three objections

Three good objections deserve honest answers.

The first is that nature has no preferred state. There is no baseline the biosphere is trying to return to. A watershed in 1750 is not more real than a watershed in 2026. This is correct. What is proposed here does not appeal to a native state. It appeals to two things a bad-faith reader cannot dismiss. Conservation of mass and energy at the boundary, which are not preferences. And whatever limits the applicable regulator has actually set, which are the record of what a democracy has agreed to call harm.

The second is that an environmental check is itself a proxy. Optimize the measurement long enough and a capable system will move the measurement without moving the underlying thing. This is serious. It is the objection the whole field is trying to answer. Two responses. At the boundary, the measurements are the reality that matters for permit and law. What crosses the fence line is what crosses the fence line. And if every disclosure, refusal, and measurement is signed and timestamped, systematic gaming leaves a signature. Not immunity from proxy failure. A shorter half-life for it.

The third is that ecological thresholds are contested. Ecologists disagree about whether a fishery is collapsed, whether a species is functionally extinct, whether a nutrient level is dangerous. At the frontier of the science that is true, and it is what a healthy science looks like. At the level where operational decisions get made—a permit limit, a watershed classification, a species listing—the science is not contested. It is written down. The thresholds a system like this would enforce are not the frontier. They are the settled record.

None of these answers is complete. If a reader can break the argument, the correspondence line is info@jedanderson.org.

There is a version of the next decade in which we ship hyper-capable autonomous agents into every corner of the physical economy. And we discover, one dry riverbed and one collapsed permit at a time, that they were brilliant at their assigned tasks and blind to everything else. The tasks will succeed. The world will pay.

There is another version. In that version, the AI safety community notices, in time, that its problem and the environmental problem are the same problem, and it builds for both. The White House writes a rulebook that names rivers. The Summer 2027 AI Safety Index has a seventh domain. Frontier labs publish environmental impact reports the same way they publish red-team results. This work is beginning in more than one place at once, and it will need every group that wants to do it.

The Lorax lost because he could only speak. We do not have that limitation.

The trees still cannot talk. But they can, at last, compute.

Sources and notes

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Revision history

- **2026-10-25.** Removed an unfilled placeholder block that had been published at the center of the essay; the surrounding sections are rejoined and the argument is otherwise unchanged. Self-citations 22–24 now use absolute

jedanderson.org URLs so they resolve inside the PDF. PDF regenerated from the corrected page.

- **2026-08-05.** Initial publication.

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